Glucose - Blerd

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Data Set Up

```r
# 1.
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3

── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   4.0.1     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.4
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b
```

```r
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```r
library(gt)
library(modelsummary)
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```r
library(modelbased)
```

```
Warning: package 'modelbased' was built under R version 4.4.3
```

```r
library(multcomp)
```

```
Loading required package: mvtnorm

Attaching package: 'mvtnorm'

The following object is masked from 'package:modelbased':

    standardize

Loading required package: survival
Loading required package: TH.data
Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select


Attaching package: 'TH.data'

The following object is masked from 'package:MASS':

    geyser
```

```r
library(ggpubr)
library(cowplot)
```

```

Attaching package: 'cowplot'

The following object is masked from 'package:ggpubr':

    get_legend

The following object is masked from 'package:gt':

    as_gtable

The following object is masked from 'package:lubridate':

    stamp
```

```r
library(lme4)
```

```
Warning: package 'lme4' was built under R version 4.4.3

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```r
library(lmerTest)
```

```

Attaching package: 'lmerTest'

The following object is masked from 'package:lme4':

    lmer

The following object is masked from 'package:stats':

    step
```

```r
library(outliers)
library(DHARMa)
```

```
This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package
```

```r
library(statmod)
library(readr)
library(dplyr)
library(EnvStats)
```

```

Attaching package: 'EnvStats'

The following object is masked from 'package:Matrix':

    print

The following object is masked from 'package:MASS':

    boxcox

The following object is masked from 'package:car':

    qqPlot

The following objects are masked from 'package:stats':

    predict, predict.lm

The following object is masked from 'package:base':

    print.default
```

```r
library(performance)
```

```
Warning: package 'performance' was built under R version 4.4.3
```

```r
library(see)
library(patchwork)
```

```

Attaching package: 'patchwork'

The following object is masked from 'package:cowplot':

    align_plots

The following object is masked from 'package:MASS':

    area
```

```r
# 2.
setwd("/Users/ben/Desktop/Blerd/Data Analysis")

# 3. Load your data
datum = read.csv("Blerd.csv")
datumlong = read.csv("BlerdData_long.csv")

# 4. Ensure categorical variables are properly set
datum$SexF = as.factor (datum$Sex)
datum$TreatmentF = as.factor(datum$Treatment)
datum$TimeF = as.factor(datum$Time)
datum$IncubatorF = as.factor (datum$Incubator)
datum$SideF = as.factor (datum$Side)
datum$PostGlucose = as.numeric (datum$PostGlucose)
```

```
Warning: NAs introduced by coercion
```

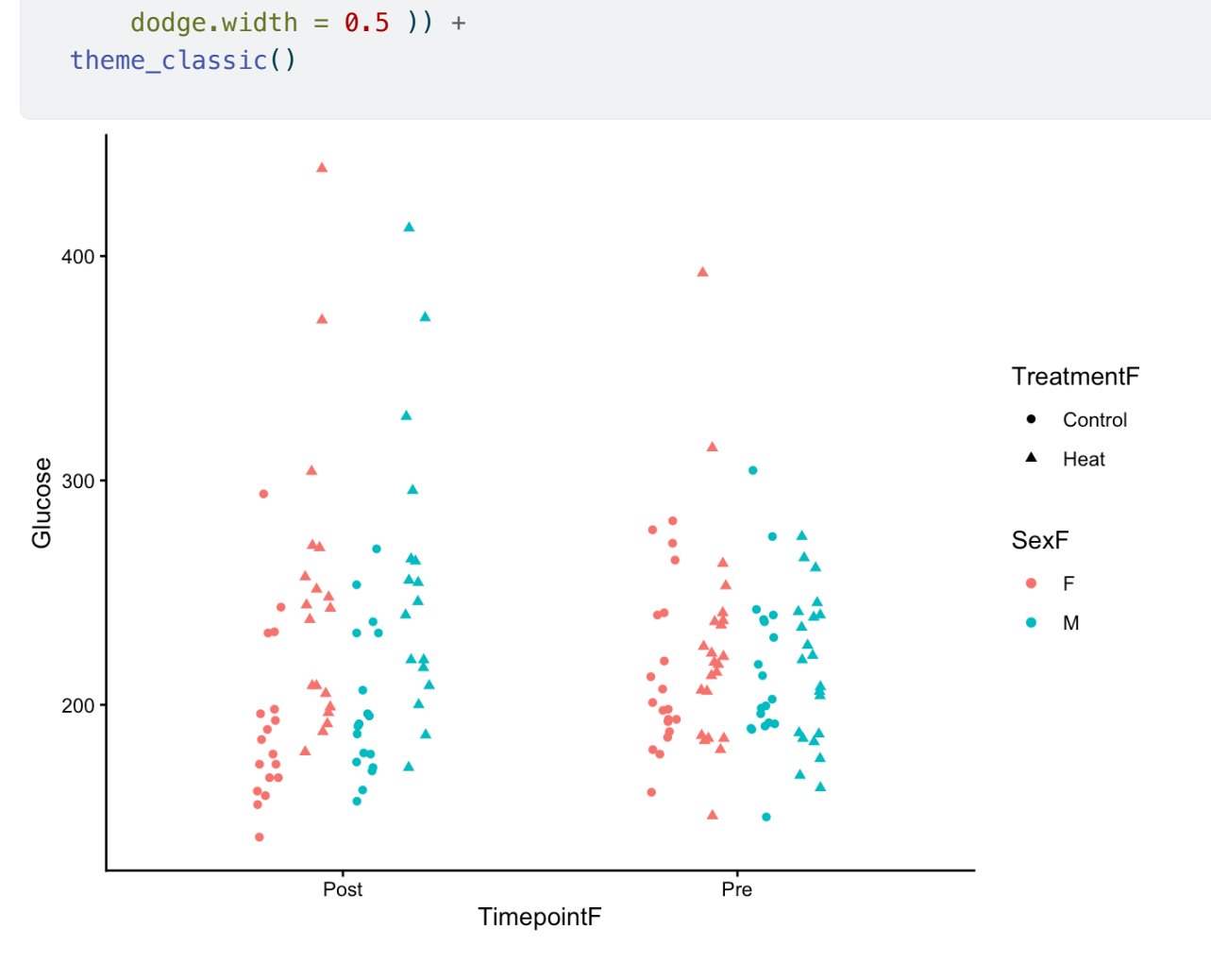
```r
datum$PreGlucose = as.numeric (datum$PreGlucose)

datumlong$SexF = as.factor (datumlong$Sex)
datumlong$TreatmentF = as.factor(datumlong$Treatment)
datumlong$TimepointF = as.factor(datumlong$Timepoint)
datumlong$Glucose = as.numeric (datumlong$Glucose)
```

```
Warning: NAs introduced by coercion
```

```r
# 5. Subset data so there are no dead individuals included
datum1 = datum %>%
  filter(!Died %in% c("Died", "PIT.Death"))

datumlong1 = datumlong %>%
  filter(!Died %in% c("Died", "PIT.Death"))

# 6. Take out any of the blank cells that cause problems when graphing
datum2 = datum1 %>%
  filter(PostGlucose != "", PostGlucose != "Dead", AvgMaxTemp != "", AvgMaxTemp != "Dead
datum2$TreatmentF = as.factor(datum2$Treatment)

datumlong2 = datumlong1 %>%
  filter(Glucose != "", Glucose != "Dead")
```

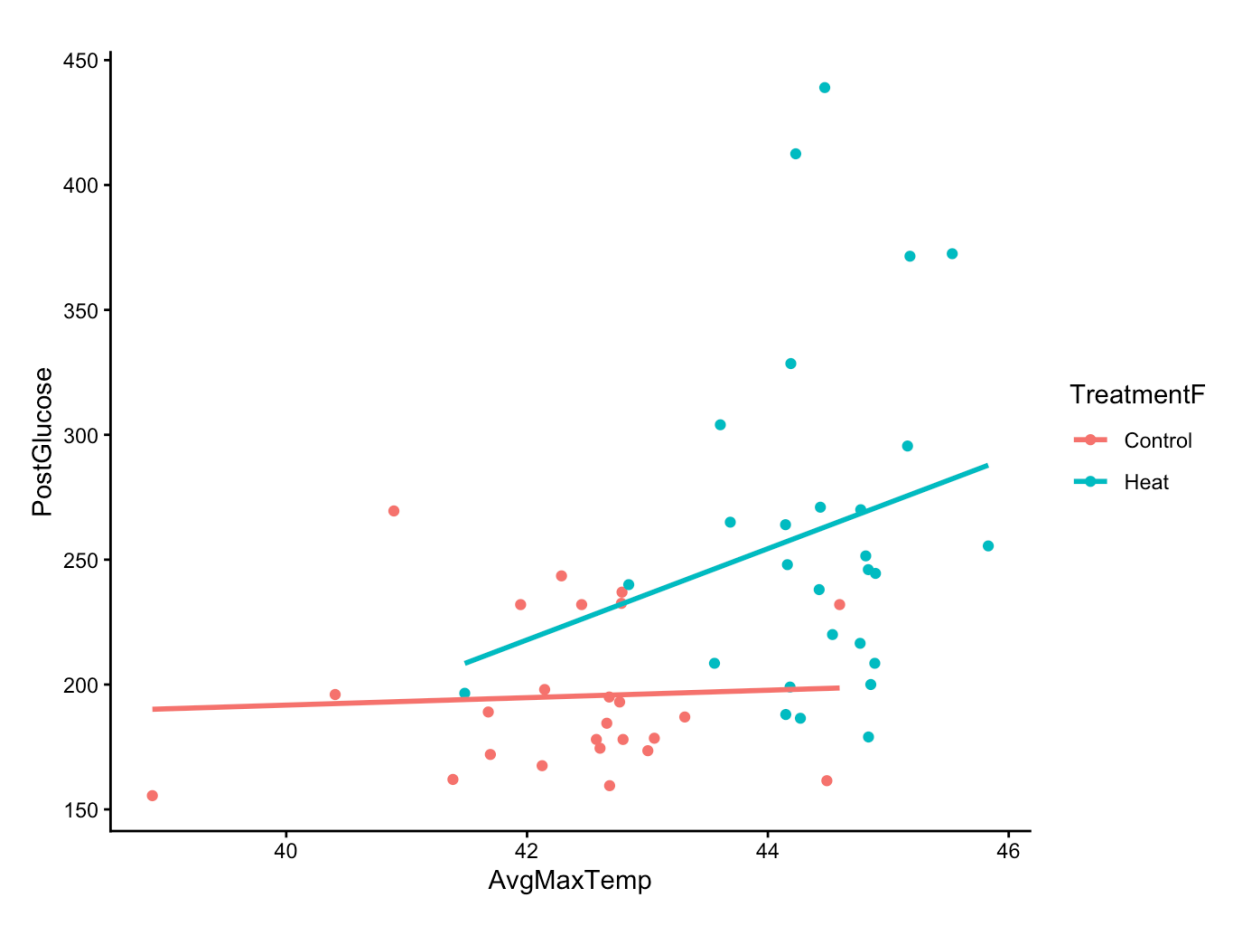
Initial Data Viz

```r
# 1. Glucose across timepoints and relationship to treatment and sex
ggplot(datumlong2, aes(x = TimepointF, y = Glucose, color = SexF, shape = TreatmentF)) +
  geom_point(position = position_jitterdodge(
      jitter.width = 0.2,
      dodge.width = 0.5 )) +
  theme_classic()
```



```r
# 2. Correlation between the average top 10% of body temperature values and post glucose
ggplot(datum2, aes(x = AvgMaxTemp, y = PostGlucose, color = TreatmentF)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Summary Stats

```r
# 1. Averages for all variables of interest
table = datumlong2 %>%
  group_by(TreatmentF, TimepointF, SexF) %>%
  summarise(
    meanGlucose = mean(Glucose, na.rm = TRUE),
    sdGlucose = sd(Glucose, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

print(table)
```

```
  TreatmentF TimepointF SexF meanGlucose sdGlucose  n
1    Control       Post    F    191.1111  37.88627 18
2    Control       Post    M    199.0556  32.58649 18
3    Control        Pre    F    214.2500  36.17847 20
4    Control        Pre    M    215.6316  35.35331 19
5       Heat       Post    F    248.0789  65.69654 19
6       Heat       Post    M    256.3235  64.62075 17
7       Heat        Pre    F    225.7536  49.52458 23
8       Heat        Pre    M    216.1429  32.96177 21
```

```r
# 2. Averages for graph
graphAvg = datumlong2 %>%
  group_by(TreatmentF, TimepointF) %>%
  summarise(
    N = n(),
    Mean_Response = mean(Glucose, na.rm = TRUE),
    SD = sd(Glucose, na.rm = TRUE),
    SE = SD / sqrt(N),
    CI_Margin = 1.96 * SE,
    CI_Upper = Mean_Response + CI_Margin,
    CI_Lower = Mean_Response - CI_Margin,
    .groups = 'drop'
  )

print(graphAvg)
```

```
  TreatmentF TimepointF  N Mean_Response       SD        SE CI_Margin CI_Upper
1    Control       Post 36      195.0833 35.05964  5.843274  11.45282 206.5362
2    Control        Pre 39      214.9231 35.31243  5.654514  11.08285 226.0059
3       Heat       Post 36      251.9722 64.38988 10.731646  21.03403 273.0062
4       Heat        Pre 44      221.1667 42.23482  6.367139  12.47959 233.6463
  CI_Lower
1 183.6305
2 203.8402
3 230.9382
4 208.6871
```

Analysis - Glucose across time points

```r
#Running the models

#  1. Start with global linear mixed effects model to assess ketone levels
results = lmer(Glucose ~ TreatmentF * SexF * TimepointF + (1|ID),
               data = datumlong2, na.action = na.omit, REML = FALSE)
summary(results)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Glucose ~ TreatmentF * SexF * TimepointF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
   1636.1    1666.5    -808.0    1616.1       145

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.0011 -0.5867 -0.1589  0.4213  4.0311

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept)  321.3   17.92
 Residual             1677.7   40.96
Number of obs: 155, groups:  ID, 83

Fixed effects:
                                 Estimate Std. Error       df t value
(Intercept)                      191.0263    10.5247 152.8090  18.150
TreatmentFHeat                    56.7649    14.6802 153.2052   3.867
SexFM                              8.5594    14.8888 152.5280   0.575
TimepointFPre                     23.2237    13.3638  79.4329   1.738
TreatmentFHeat:SexFM              -0.6328    21.0577 153.1375  -0.030
TreatmentFHeat:TimepointFPre     -45.2613    18.5015  81.0112  -2.446
SexFM:TimepointFPre               -7.1778    18.9971  78.4361  -0.378
TreatmentFHeat:SexFM:TimepointFPre -10.3595    26.5746  80.7248  -0.390
                                 Pr(>|t|)
(Intercept)                       < 2e-16 ***
TreatmentFHeat                   0.000163 ***
SexFM                            0.566214
TimepointFPre                    0.086121 .
TreatmentFHeat:SexFM             0.976065
TreatmentFHeat:TimepointFPre     0.016596 *
SexFM:TimepointFPre              0.706572
TreatmentFHeat:SexFM:TimepointFPre 0.697690
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TrFH:SFM TFH:TF SFM:TF
TreatmntFHt -0.717
SexFM       -0.707  0.507
TimepontFPr -0.673  0.483  0.476
TrtmnFH:SFM  0.500 -0.697 -0.707 -0.337
TrtmnFH:TFP  0.486 -0.683 -0.344 -0.722  0.476
SxFM:TmpnFP  0.474 -0.340 -0.667 -0.703  0.472    0.508
TFH:SFM:TFP -0.339  0.475  0.477  0.503 -0.681   -0.696 -0.715
```

```r
Anova(results, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Glucose
                            Chisq Df Pr(>Chisq)
(Intercept)              329.4302  1  < 2.2e-16 ***
TreatmentF                14.9518  1  0.0001103 ***
SexF                       0.3305  1  0.5653672
TimepointF                 3.0200  1  0.0822453 .
TreatmentF:SexF            0.0009  1  0.9760254
TreatmentF:TimepointF      5.9847  1  0.0144309 *
SexF:TimepointF            0.1428  1  0.7055514
TreatmentF:SexF:TimepointF   0.1520  1  0.6966636
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 2. The three way interaction was not significant, so it was removed and all two way in
results2 = lmer(Glucose ~ TreatmentF * SexF + TreatmentF * TimepointF +
                 SexF * TimepointF + (1|ID), data = datumlong2, na.action = na.omit,
summary(results2)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Glucose ~ TreatmentF * SexF + TreatmentF * TimepointF + SexF *
    TimepointF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
   1634.2    1661.6    -808.1    1616.2       146

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.9734 -0.6054 -0.1477  0.4080  3.9973

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept)  321.9   17.94
 Residual             1679.1   40.98
Number of obs: 155, groups:  ID, 83

Fixed effects:
                             Estimate Std. Error      df t value Pr(>|t|)
(Intercept)                   189.637      9.908 145.741  19.140  < 2e-16 ***
TreatmentFHeat                 59.485     12.922 135.173   4.604 9.47e-06 ***
SexFM                          11.328     13.093 129.521   0.865 0.388549
TimepointFPre                  25.843     11.556  79.824   2.236 0.028118 *
TreatmentFHeat:SexFM           -6.225     15.425  82.572  -0.404 0.687589
TreatmentFHeat:TimepointFPre  -50.282     13.287  80.698  -3.784 0.000295 ***
SexFM:TimepointFPre           -12.472     13.290  80.786  -0.938 0.350802
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TFH:SF TFH:TF
TreatmntFHt -0.672
SexFM       -0.659  0.362
TimepontFPr -0.618  0.320  0.311
TrtmnFH:SFM  0.391 -0.580 -0.594  0.009
TrtmnFH:TFP  0.371 -0.557 -0.019 -0.600  0.003
SxFM:TmpnFP  0.352  0.000 -0.531 -0.569 -0.030  0.021
```

```r
Anova(results2, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Glucose
                       Chisq Df Pr(>Chisq)
(Intercept)           366.3275  1  < 2.2e-16 ***
TreatmentF             21.1925  1  4.154e-06 ***
SexF                    0.7485  1  0.3869490
TimepointF              5.0014  1  0.0253275 *
TreatmentF:SexF         0.1629  1  0.6865455
TreatmentF:TimepointF  14.3217  1  0.0001541 ***
SexF:TimepointF         0.8807  1  0.3480045
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 3. The Treatment*Sex interaction was not significant, so it was removed from the model
results3 = lmer(Glucose ~ SexF * TimepointF + TreatmentF * TimepointF +
                 (1|ID), data = datumlong2, na.action = na.omit, REML = FALSE)
summary(results3)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Glucose ~ SexF * TimepointF + TreatmentF * TimepointF + (1 |      ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
   1632.4    1656.7    -808.2    1616.4       147

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.9479 -0.5922 -0.1636  0.4183  4.0212

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept)  322.2   17.95
 Residual             1680.9   41.00
Number of obs: 155, groups:  ID, 83

Fixed effects:
                             Estimate Std. Error      df t value Pr(>|t|)
(Intercept)                   191.199      9.125 152.915  20.953  < 2e-16 ***
SexFM                           8.190     10.539 153.162   0.777 0.438322
TimepointFPre                  25.888     11.562  79.752   2.239 0.027935 *
TreatmentFHeat                 56.463     10.535 153.133   5.359 3.02e-07 ***
SexFM:TimepointFPre           -12.631     13.291  80.744  -0.950 0.344753
TimepointFPre:TreatmentFHeat  -50.264     13.294  80.623  -3.781 0.000298 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) SexFM  TmpnFP TrtmFH SFM:TF
SexFM       -0.577
TimepontFPr -0.676  0.393
TreatmntFHt -0.593  0.027  0.400
SxFM:TmpnFP  0.395 -0.682 -0.569 -0.021
TmpntFP:TFH  0.402 -0.021 -0.600 -0.681  0.021
```

```r
Anova(results3, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Glucose
                       Chisq Df Pr(>Chisq)
(Intercept)           439.0270  1  < 2.2e-16 ***
SexF                    0.6038  1  0.4371235
TimepointF              5.0136  1  0.0251487 *
TreatmentF             28.7224  1  8.353e-08 ***
SexF:TimepointF         0.9032  1  0.3419170
TimepointF:TreatmentF  14.2961  1  0.0001562 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 4. The Sex*Timepoint interaction was not significant, so it was removed from the model
results4 = lmer(Glucose ~ TreatmentF * SexF + TimepointF + (1|ID), data = datumlong2,
               na.action = na.omit, REML = FALSE)
summary(results4)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Glucose ~ TreatmentF * SexF + TimepointF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
   1644.2    1665.5    -815.1    1630.2       148

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.9437 -0.6302 -0.1866  0.4027  4.1496

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept)  189.8   13.78
 Residual             1981.1   44.51
Number of obs: 155, groups:  ID, 83

Fixed effects:
                     Estimate Std. Error      df t value Pr(>|t|)
(Intercept)           206.506      8.735 111.933  23.642  < 2e-16 ***
TreatmentFHeat         32.520     10.845  81.671   2.999  0.00359 **
SexFM                   4.325     11.203  78.978   0.386  0.70047
TimepointFPre          -6.032      7.194  80.458  -0.838  0.40426
TreatmentFHeat:SexFM   -6.184     15.583  81.352  -0.397  0.69251
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP
TreatmntFHt -0.646
SexFM       -0.636  0.509
TimepontFPr -0.435 -0.015  0.009
TrtmnFH:SFM  0.458 -0.696 -0.719 -0.009
```

```r
Anova(results4, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Glucose
                  Chisq Df Pr(>Chisq)
(Intercept)     558.9449  1  < 2.2e-16 ***
TreatmentF        8.9916  1   0.002712 **
SexF              0.1491  1   0.699434
TimepointF        0.7030  1   0.401775
TreatmentF:SexF   0.1575  1   0.691471
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 5. The Sex*Treatment interaction was not significant, so it was removed from the model
results5 = lmer(Glucose ~ TreatmentF + SexF + TimepointF + (1|ID), data = datumlong2,
               na.action = na.omit, REML = FALSE)
summary(results5)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Glucose ~ TreatmentF + SexF + TimepointF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
   1642.3    1660.6    -815.2    1630.3       149

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.9155 -0.6297 -0.2012  0.3842  4.1741

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept)  189.8   13.78
 Residual             1983.3   44.53
Number of obs: 155, groups:  ID, 83

Fixed effects:
               Estimate Std. Error      df t value Pr(>|t|)
(Intercept)     208.094      7.768 121.140  26.790  < 2e-16 ***
TreatmentFHeat   29.526      7.794  81.320   3.788  0.00029 ***
SexFM             1.129      7.790  81.447   0.145  0.88515
TimepointFPre    -6.057      7.198  80.389  -0.842  0.40253
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM
TreatmntFHt -0.513
SexFM       -0.496  0.017
TimepontFPr -0.485 -0.030  0.004
```

```r
Anova(results5, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Glucose
               Chisq Df Pr(>Chisq)
(Intercept) 717.7131  1  < 2.2e-16 ***
TreatmentF   14.3506  1  0.0001517 ***
SexF          0.0210  1  0.8847877
TimepointF    0.7082  1  0.4000370
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Use AIC to test which one of these models has the best fit
AIC(results, results2, results3, results4, results5)
```

```
         df      AIC
results  10 1636.051
results2  9 1634.203
```

```
results3  8 1632.366
results4  7 1644.168
results5  6 1642.326
```

```r
#This is the global model, use 'results3' for testing the assumptions

results_final2 = lmer(Glucose ~ SexF * TimepointF + TreatmentF * TimepointF +
                   (1|ID), data = datumlong2, na.action = na.omit, REML = TRUE)
summary(results_final2)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Glucose ~ SexF * TimepointF + TreatmentF * TimepointF + (1 |      ID)
   Data: datumlong2

REML criterion at convergence: 1580.2

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-1.9089 -0.5809 -0.1606  0.4103  3.9432 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept)  334.2   18.28   
 Residual             1749.4   41.83   
Number of obs: 155, groups:  ID, 83

Fixed effects:
                             Estimate Std. Error      df t value Pr(>|t|)    
(Intercept)                   191.199      9.307 147.029  20.544  < 2e-16 ***
SexFM                           8.189     10.749 147.263   0.762 0.447359    
TimepointFPre                  25.888     11.795  76.739   2.195 0.031195 *  
TreatmentFHeat                 56.465     10.745 147.236   5.255  5.1e-07 ***
SexFM:TimepointFPre           -12.631     13.559  77.692  -0.932 0.354440    
TimepointFPre:TreatmentFHeat  -50.265     13.562  77.576  -3.706 0.000392 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) SexFM  TmpnFP TrtmFH SFM:TF
SexFM       -0.577                            
TimepontFPr -0.676  0.393                     
TreatmntFHt -0.593  0.027  0.400              
SxFM:TmpnFP  0.395 -0.682 -0.569 -0.021       
TmpntFP:TFH  0.402 -0.021 -0.600 -0.681  0.021
```

```r
Anova(results_final2, type = "III")
```

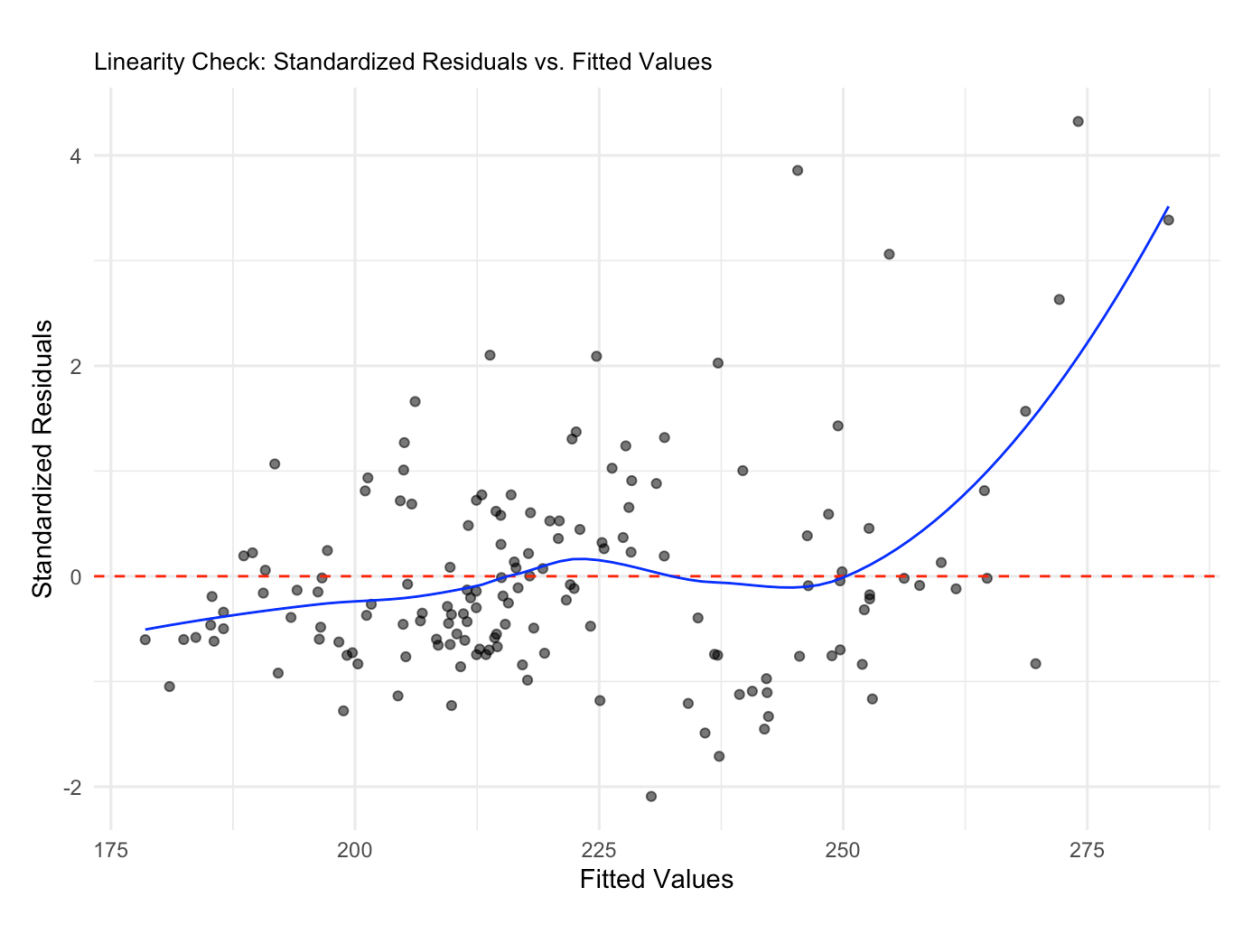
```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Glucose
                        Chisq Df Pr(>Chisq)    
(Intercept)          422.0532  1  < 2.2e-16 ***
SexF                   0.5804  1  0.4461403    
TimepointF             4.8175  1  0.0281716 *  
TreatmentF            27.6138  1  1.481e-07 ***
SexF:TimepointF        0.8678  1  0.3515537    
TimepointF:TreatmentF 13.7376  1  0.0002102 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Assumptions of the model


# 1. Check the VIF of this model
vif(results_final2)
```

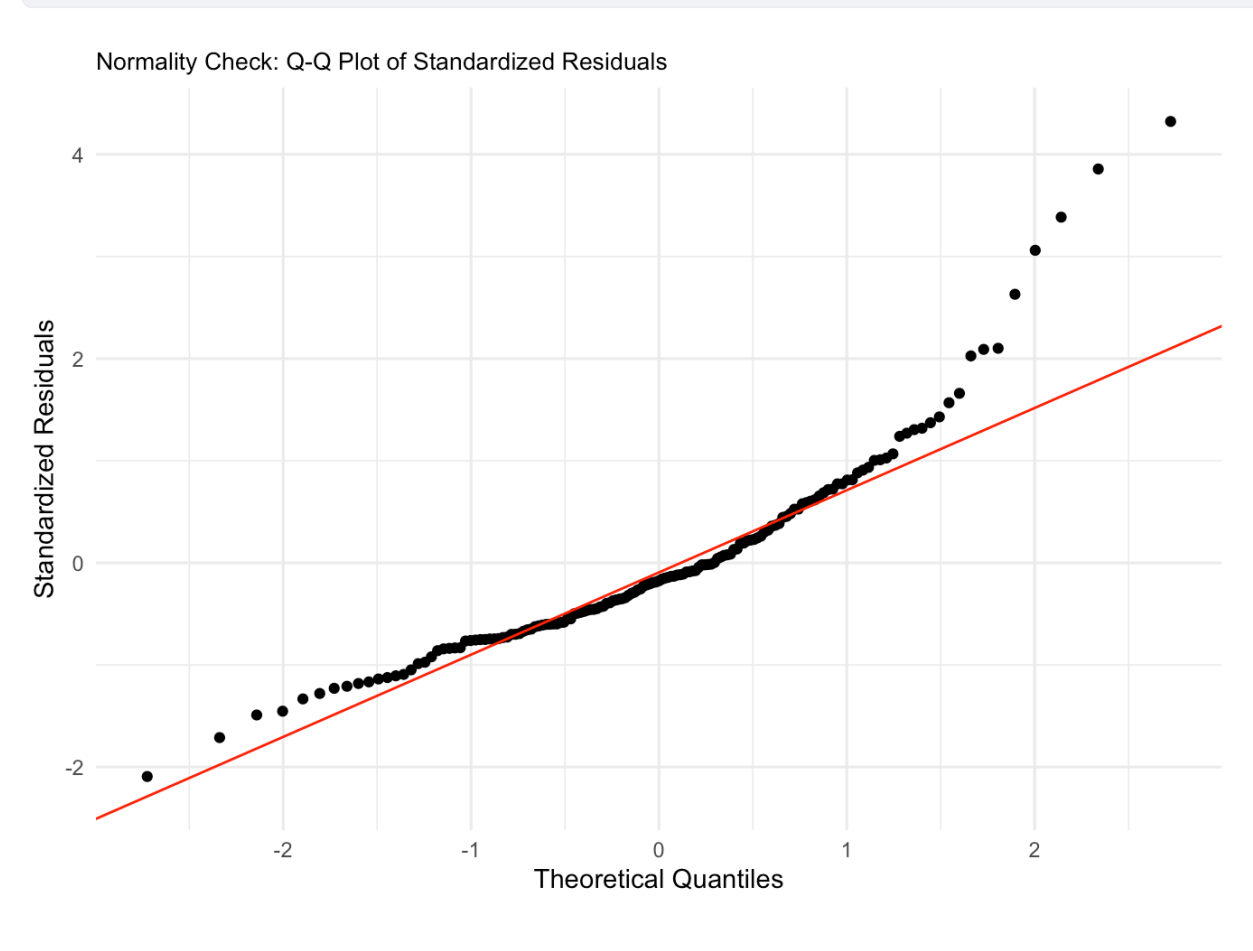
```
                 SexF            TimepointF            TreatmentF 
             1.870791              3.031381              1.868942 
      SexF:TimepointF TimepointF:TreatmentF 
             2.803297              2.972706 
```

```r
  #VIF = 1: No multicollinearity
  #VIF between 1 and 5: Generally considered acceptable or moderate multicollinearity
  #VIF > 5 (or > 10): Indicates potentially high multicollinearity

# 2. Check other assumptions of a linear model

# --- Extract Residuals and Fitted Values ---
#1. Extract the data frame used by the model
datum_clean <- model.frame(results_final2)

# 2. Calculate and assign the fitted and residual values to this clean data frame
datum_clean$fitted <- fitted(results_final2)
datum_clean$raw_resid <- residuals(results_final2)
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)

# 3. Use the studentized residuals method from the 'lme4' utility functions
# Although 'rstandard' doesn't work, 'residuals' is robust.
# If you really need a standardized residual, use this function:
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)


# --- 2. Create the Plots ---

# A. Linearity and Homoscedasticity Plot (Residuals vs. Fitted)
ggplot(datum_clean, aes(x = fitted, y = std_resid)) +
  geom_point(alpha = 0.6) +
  # Add the ideal reference line at y=0
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  # Add a LOESS smoothing line (red line in performance)
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Linearity Check: Standardized Residuals vs. Fitted Values",
    x = "Fitted Values",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

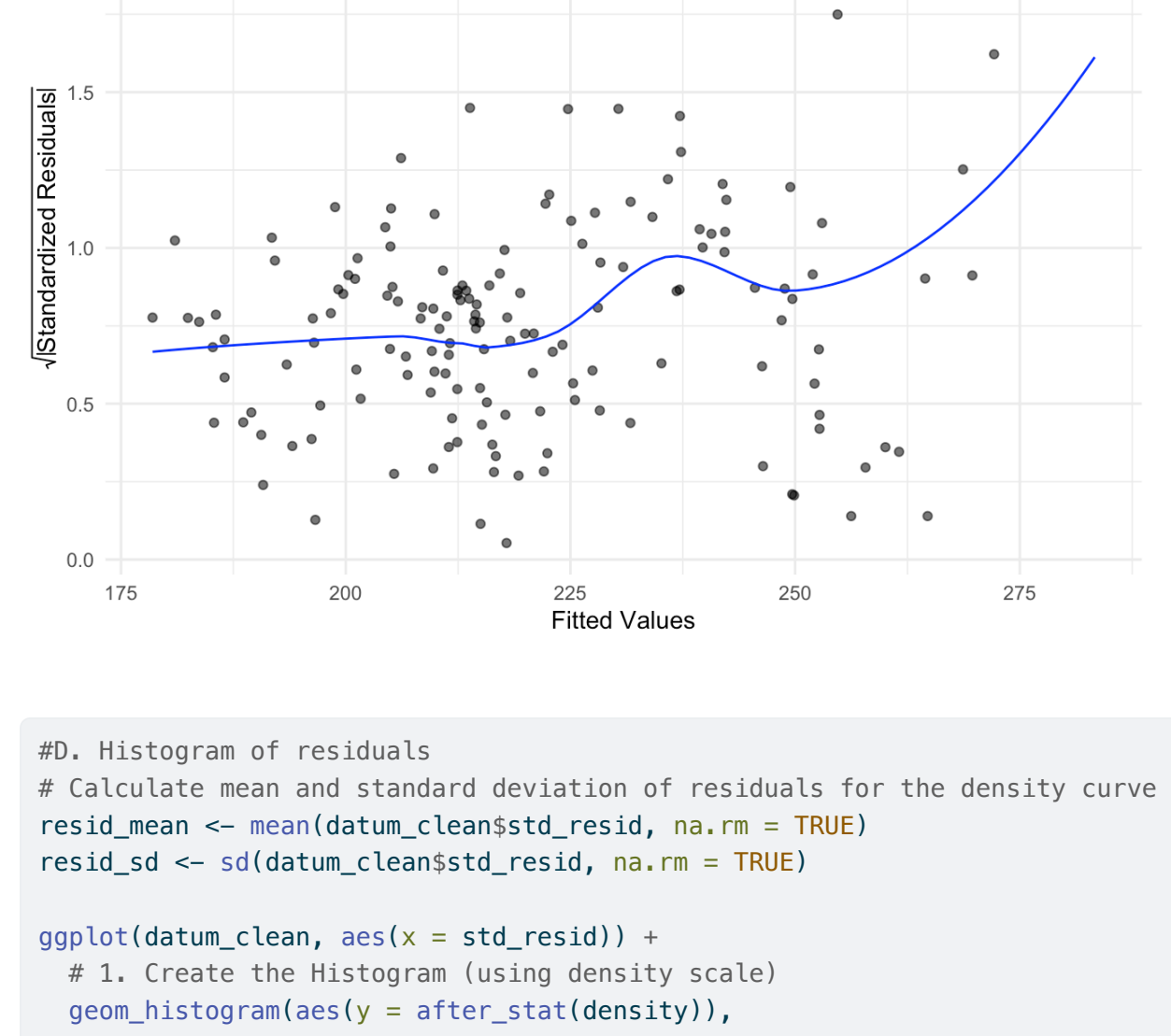


```r
# B. Normality Check (Quantile-Quantile Plot)
ggplot(datum_clean, aes(sample = std_resid)) +
  # Create the Q-Q plot
  stat_qq() +
  # Add the reference line (the ideal normal distribution line)
  stat_qq_line(color = "red") +
  labs(
    title = "Normality Check: Q-Q Plot of Standardized Residuals",
    x = "Theoretical Quantiles",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

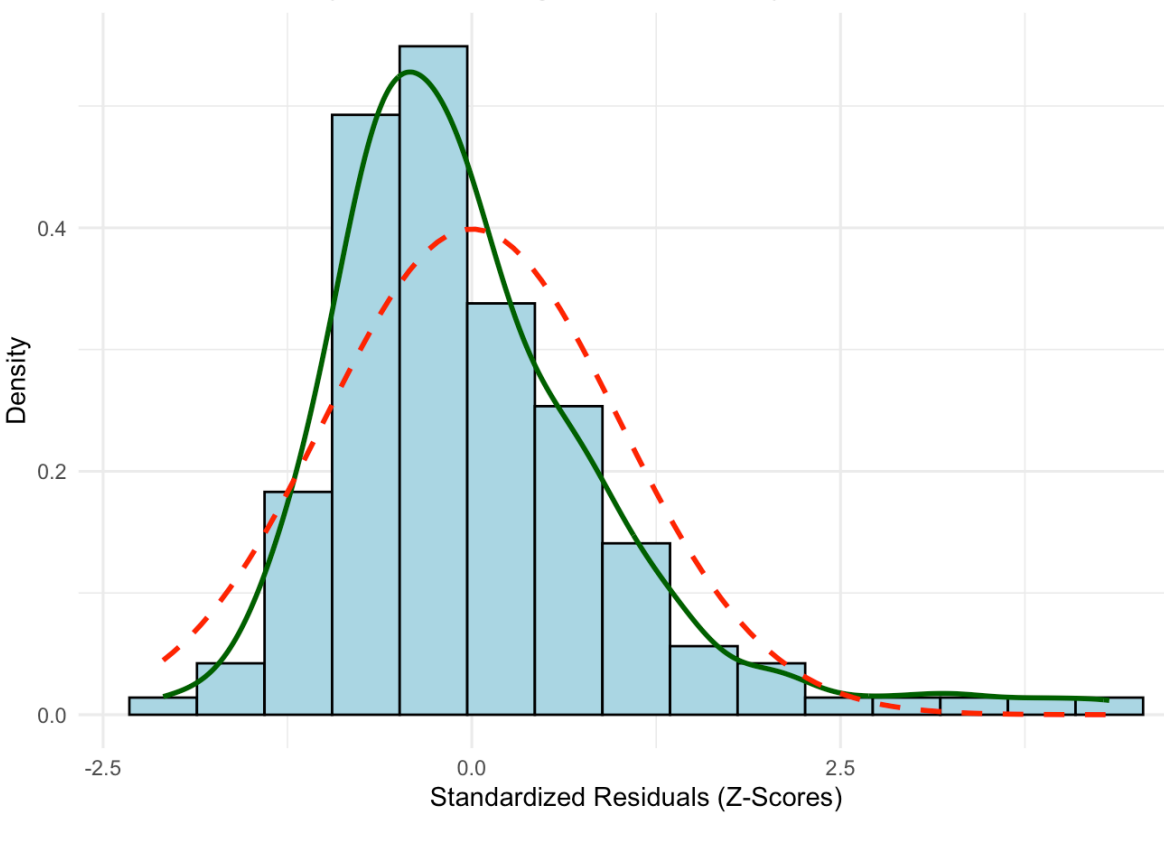


```r
# C. Scale-Location Plot (Homogeneity of Variance Check)
# This uses the square root of the absolute standardized residuals
ggplot(datum_clean, aes(x = fitted, y = sqrt(abs(std_resid)))) +
  geom_point(alpha = 0.6) +
  # Add the ideal flat reference line
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Homogeneity of Variance Check: Scale-Location Plot",
    x = "Fitted Values",
    y = expression(sqrt(abs("Standardized Residuals")))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```r
#D. Histogram of residuals
# Calculate mean and standard deviation of residuals for the density curve
resid_mean <- mean(datum_clean$std_resid, na.rm = TRUE)
resid_sd <- sd(datum_clean$std_resid, na.rm = TRUE)

ggplot(datum_clean, aes(x = std_resid)) +
  # 1. Create the Histogram (using density scale)
  geom_histogram(aes(y = after_stat(density)),
                 bins = 15, # Adjust bin count for best visualization
                 color = "black",
                 fill = "lightblue") +

  # 2. Overlay the Density of the Residuals (Green Line)
  geom_density(color = "darkgreen", linewidth = 1) +

  # 3. Overlay the Theoretical Normal Density Curve (Red Dashed Line)
  stat_function(fun = dnorm,
                args = list(mean = resid_mean, sd = resid_sd),
                color = "red",
                linewidth = 1,
                linetype = "dashed") +

  labs(
    title = "Residual Normality Check: Histogram with Density Curve",
    x = "Standardized Residuals (Z-Scores)",
    y = "Density"
  ) +
  theme_minimal()
```



```r
emms_interaction = emmeans(results_final2, ~ TreatmentF * TimepointF)
interaction_pairs = pairs(emms_interaction, adjust = "sidak")
print(interaction_pairs)
```

```
 contrast                   estimate   SE    df t.ratio p.value
 Control Post - Heat Post      -56.5 10.8 147.2  -5.245  <.0001
 Control Post - Control Pre    -19.6  9.7  73.7  -2.017  0.2526
 Control Post - Heat Pre       -25.8 10.3 146.1  -2.512  0.0761
 Heat Post - Control Pre        36.9 10.6 146.8   3.495  0.0038
 Heat Post - Heat Pre           30.7  9.5  78.3   3.232  0.0108
 Control Pre - Heat Pre         -6.2 10.0 145.5  -0.618  0.9903

Results are averaged over the levels of: SexF 
Degrees-of-freedom method: kenward-roger 
P value adjustment: sidak method for 6 tests 
```

```r
cld_output = cld(emms_interaction,
                 adjust = "sidak",
                 Letters = letters)


print(cld_output)
```

```
  TreatmentF TimepointF   emmean       SE       df lower.CL upper.CL .group
1    Control       Post 195.2935 7.608332 146.5500 176.1062 214.4809     a 
3    Control        Pre 214.8661 7.310493 145.5389 196.4283 233.3040     a 
4       Heat        Pre 221.0657 6.885310 145.5389 203.7002 238.4312     a 
2       Heat       Post 251.7585 7.614907 147.6722 232.5564 270.9606      b
```

Conclusions: Based on the residuals from the model, this fit **does not meet the assumptions**.

Next steps: Try log transforming the data

```r
#Log-transform the data
datumlong2$log_Glucose = log(datumlong2$Glucose)

#Running the models

#  1. Start with global linear mixed effects model to assess ketone levels
results = lmer(log_Glucose ~ TreatmentF * SexF * TimepointF + (1|ID),
               data = datumlong2, na.action = na.omit, REML = FALSE)
summary(results)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: log_Glucose ~ TreatmentF * SexF * TimepointF + (1 | ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid 
   -71.3    -40.9     45.6     -91.3       145 

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-2.4430 -0.6431 -0.1033  0.5119  3.0481 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.007193 0.08481 
 Residual             0.026029 0.16134 
Number of obs: 155, groups:  ID, 83

Fixed effects:
                                    Estimate Std. Error        df t value
(Intercept)                          5.23565    0.04286 150.92277 122.155
TreatmentFHeat                       0.24899    0.05976 151.63725   4.166
SexFM                                0.04902    0.06065 150.41924   0.808
TimepointFPre                        0.11861    0.05271  77.96197   2.250
TreatmentFHeat:SexFM                -0.01855    0.08573 151.51477  -0.216
TreatmentFHeat:TimepointFPre        -0.20354    0.07302  79.48817  -2.787
SexFM:TimepointFPre                 -0.04194    0.07491  76.99630  -0.560
TreatmentFHeat:SexFM:TimepointFPre  -0.02354    0.10487  79.21148  -0.224
                                   Pr(>|t|)    
(Intercept)                         < 2e-16 ***
TreatmentFHeat                     5.18e-05 ***
SexFM                               0.42017    
TimepointFPre                       0.02727 *  
TreatmentFHeat:SexFM                0.82900    
TreatmentFHeat:TimepointFPre        0.00664 ** 
SexFM:TimepointFPre                 0.57718    
TreatmentFHeat:SexFM:TimepointFPre  0.82297    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TrFH:SFM TFH:TF SFM:TF
TreatmntFHt -0.717                                            
SexFM       -0.707  0.507                                     
TimepontFPr -0.654  0.469  0.462                              
TrtmnFH:SFM  0.500 -0.697 -0.707 -0.327                       
TrtmnFH:TFP  0.472 -0.664 -0.334 -0.722  0.463                
SxFM:TmpnFP  0.460 -0.330 -0.647 -0.704  0.458    0.508       
TFH:SFM:TFP -0.329  0.463  0.462  0.503 -0.662   -0.696 -0.714
```

```r
Anova(results, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: log_Glucose
                                Chisq Df Pr(>Chisq)    
(Intercept)                14921.8992  1  < 2.2e-16 ***
TreatmentF                    17.3590  1  3.094e-05 ***
SexF                           0.6534  1   0.418896    
TimepointF                     5.0625  1   0.024450 *  
TreatmentF:SexF                0.0468  1   0.828711    
TreatmentF:TimepointF          7.7694  1   0.005314 ** 
SexF:TimepointF                0.3135  1   0.575553    
TreatmentF:SexF:TimepointF     0.0504  1   0.822391    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 2. The three way interaction was not significant, so it was removed and all two way in
results2 = lmer(log_Glucose ~ TreatmentF * SexF + TreatmentF * TimepointF +
                SexF * TimepointF + (1|ID), data = datumlong2, na.action = na.omit,
                REML = FALSE)
summary(results2)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: log_Glucose ~ TreatmentF * SexF + TreatmentF * TimepointF + SexF *
    TimepointF + (1 | ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid 
   -73.2    -45.8     45.6     -91.2       146 

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-2.4277 -0.6386 -0.1176  0.4938  3.0290 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.007213 0.08493 
 Residual             0.026023 0.16132 
Number of obs: 155, groups:  ID, 83

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    5.23249    0.04049 142.50735 129.232  < 2e-16
TreatmentFHeat                 0.25519    0.05300 131.57557   4.815 3.98e-06
SexFM                          0.05532    0.05380 125.54276   1.028  0.30576
TimepointFPre                  0.12455    0.04557  78.37334   2.733  0.00775
TreatmentFHeat:SexFM          -0.03130    0.06425  81.75191  -0.487  0.62740
TreatmentFHeat:TimepointFPre  -0.21494    0.05241  79.21862  -4.101 9.88e-05
SexFM:TimepointFPre           -0.05395    0.05242  79.29583  -1.029  0.30651
                                 
(Intercept)                  ***
TreatmentFHeat               ***
SexFM                           
TimepointFPre                **  
TreatmentFHeat:SexFM            
TreatmentFHeat:TimepointFPre ***
SexFM:TimepointFPre             
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TFH:SF TFH:TF
TreatmntFHt -0.675                                   
SexFM       -0.662  0.373                            
TimepontFPr -0.598  0.308  0.300                     
TrtmnFH:SFM  0.399 -0.588 -0.604  0.009              
TrtmnFH:TFP  0.359 -0.538 -0.018 -0.599  0.003       
SxFM:TmpnFP  0.341  0.001 -0.511 -0.570 -0.029  0.021
```

```r
Anova(results2, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: log_Glucose
                           Chisq Df Pr(>Chisq)    
(Intercept)           16700.9285  1  < 2.2e-16 ***
TreatmentF               23.1846  1  1.472e-06 ***
SexF                      1.0575  1   0.303785    
TimepointF                7.4714  1   0.006269 ** 
TreatmentF:SexF           0.2374  1   0.626091    
TreatmentF:TimepointF    16.8208  1  4.108e-05 ***
SexF:TimepointF           1.0593  1   0.303385    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 3. The Treatment*Sex interaction was not significant, so it was removed from the model
results3 = lmer(log_Glucose ~ SexF * TimepointF + TreatmentF * TimepointF +
                  (1|ID), data = datumlong2, na.action = na.omit, REML = FALSE)
summary(results3)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: log_Glucose ~ SexF * TimepointF + TreatmentF * TimepointF + (1 |
    ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid 
   -75.0    -50.7     45.5     -91.0       147 

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-2.3982 -0.6292 -0.1206  0.4931  3.0562 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.007225 0.0850  
 Residual             0.026063 0.1614  
Number of obs: 155, groups:  ID, 83

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    5.24036    0.03716 151.10036 141.040  < 2e-16
SexFM                          0.03949    0.04290 151.54676   0.920 0.358841
TimepointFPre                  0.12476    0.04560  78.26217   2.736 0.007692
TreatmentFHeat                 0.23999    0.04289 151.49456   5.596    1e-07
SexFM:TimepointFPre           -0.05470    0.05244  79.22044  -1.043 0.300055
TimepointFPre:TreatmentFHeat  -0.21485    0.05245  79.10410  -4.097 0.000101
                                 
(Intercept)                  ***
SexFM                           
TimepointFPre                **  
TreatmentFHeat               ***
SexFM:TimepointFPre             
TimepointFPre:TreatmentFHeat ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) SexFM  TmpnFP TrtmFH SFM:TF
SexFM       -0.577                            
TimepontFPr -0.657  0.383                     
TreatmntFHt -0.594  0.027  0.388              
SxFM:TmpnFP  0.385 -0.663 -0.570 -0.021       
TmpntFP:TFH  0.390 -0.021 -0.599 -0.662  0.021
```

```r
Anova(results3, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: log_Glucose
                           Chisq Df Pr(>Chisq)    
(Intercept)           19892.2979  1  < 2.2e-16 ***
SexF                      0.8471  1   0.357378    
TimepointF                7.4857  1   0.006219 ** 
TreatmentF               31.3127  1  2.196e-08 ***
SexF:TimepointF           1.0881  1   0.296882    
TimepointF:TreatmentF    16.7820  1  4.193e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 4. The Sex*Timepoint interaction was not significant, so it was removed from the model
results4 = lmer(log_Glucose ~ TreatmentF * SexF + TimepointF + (1|ID), data = datumlong2
                na.action = na.omit, REML = FALSE)
summary(results4)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: log_Glucose ~ TreatmentF * SexF + TimepointF + (1 | ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid 
   -61.0    -39.7     37.5     -75.0       148 

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-2.3898 -0.5936 -0.1525  0.5147  3.1996 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004764 0.06902 
 Residual             0.031609 0.17779 
Number of obs: 155, groups:  ID, 83

Fixed effects:
                      Estimate Std. Error        df t value Pr(>|t|)    
(Intercept)            5.30488    0.03616 109.15883 146.713  < 2e-16 ***
TreatmentFHeat         0.13958    0.04520  80.74647   3.088  0.00276 ** 
SexFM                  0.02486    0.04672  78.08466   0.532  0.59617    
TimepointFPre         -0.01193    0.02878  79.01663  -0.415  0.67960    
TreatmentFHeat:SexFM  -0.03084    0.06495  80.42634  -0.475  0.63623    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP
TreatmntFHt -0.652                     
SexFM       -0.640  0.509              
TimepontFPr -0.421 -0.015  0.009       
TrtmnFH:SFM  0.462 -0.696 -0.719 -0.009
```

```r
Anova(results4, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: log_Glucose
                     Chisq Df Pr(>Chisq)    
(Intercept)     21524.7427  1  < 2.2e-16 ***
TreatmentF          9.5368  1   0.002014 ** 
SexF                0.2831  1   0.594657    
TimepointF          0.1718  1   0.678477    
TreatmentF:SexF     0.2254  1   0.634945    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 5. The Sex*Treatment interaction was not significant, so it was removed from the model
results5 = lmer(log_Glucose ~ TreatmentF + SexF + TimepointF + (1|ID), data = datumlong2
                na.action = na.omit, REML = FALSE)
summary(results5)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: log_Glucose ~ TreatmentF + SexF + TimepointF + (1 | ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid 
   -62.8    -44.5     37.4     -74.8       149 

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-2.3570 -0.6005 -0.1537  0.5071  3.2276 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004766 0.06904 
 Residual             0.031659 0.17793 
Number of obs: 155, groups:  ID, 83

Fixed effects:
                Estimate Std. Error         df t value Pr(>|t|)    
(Intercept)     5.312797   0.032099 118.043372 165.510  < 2e-16 ***
TreatmentFHeat  0.124658   0.032494  80.363428   3.836 0.000247 ***
SexFM           0.008903   0.032476  80.488345   0.274 0.784676    
TimepointFPre  -0.012051   0.028801  78.911413  -0.418 0.676767    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM 
TreatmntFHt -0.519              
SexFM       -0.500  0.017       
TimepontFPr -0.471 -0.029  0.004
```

```r
Anova(results5, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: log_Glucose
                 Chisq Df Pr(>Chisq)    
(Intercept) 27393.6925  1  < 2.2e-16 ***
TreatmentF     14.7177  1  0.0001249 ***
SexF            0.0752  1  0.7839735    
TimepointF      0.1751  1  0.6756301    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Use AIC to test which one of these models has the best fit
AIC(results, results2, results3, results4, results5)
```

```
         df       AIC
results  10 -71.28843
results2  9 -73.23807
results3  8 -75.00086
results4  7 -61.03320
results5  6 -62.80795
```

```r
#This is the global model, use 'results3' for testing the assumptions
results_final = lmer(log_Glucose ~ SexF * TimepointF + TreatmentF * TimepointF +
                    (1|ID), data = datumlong2, na.action = na.omit, REML = TRUE)
summary(results_final)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: log_Glucose ~ SexF * TimepointF + TreatmentF * TimepointF + (1 |
    ID)
   Data: datumlong2

REML criterion at convergence: -61.1

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-2.3504 -0.6173 -0.1179  0.4838  2.9970 

Random effects:
```

```
Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.007493 0.08656
 Residual             0.027130 0.16471
Number of obs: 155, groups:  ID, 83

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    5.24036    0.03789 145.31202 138.290  < 2e-16
SexFM                          0.03949    0.04376 145.73611   0.902 0.368318
TimepointFPre                  0.12476    0.04652  75.29516   2.682 0.008996
TreatmentFHeat                 0.24001    0.04374 145.68653   5.487 1.76e-07
SexFM:TimepointFPre           -0.05470    0.05350  76.21611  -1.022 0.309809
TimepointFPre:TreatmentFHeat  -0.21486    0.05351  76.10430  -4.015 0.000138

(Intercept)                  ***
SexFM
TimepointFPre                **
TreatmentFHeat               ***
SexFM:TimepointFPre
TimepointFPre:TreatmentFHeat ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) SexFM  TmpnFP TrtmFH SFM:TF
SexFM       -0.577
TimepontFPr -0.657  0.383
TreatmntFHt -0.594  0.027  0.388
SxFM:TmpnFP  0.385 -0.663 -0.570 -0.021
TmpntFP:TFH  0.390 -0.021 -0.599 -0.663  0.021
```

```
Anova(results_final, type = "III")
```

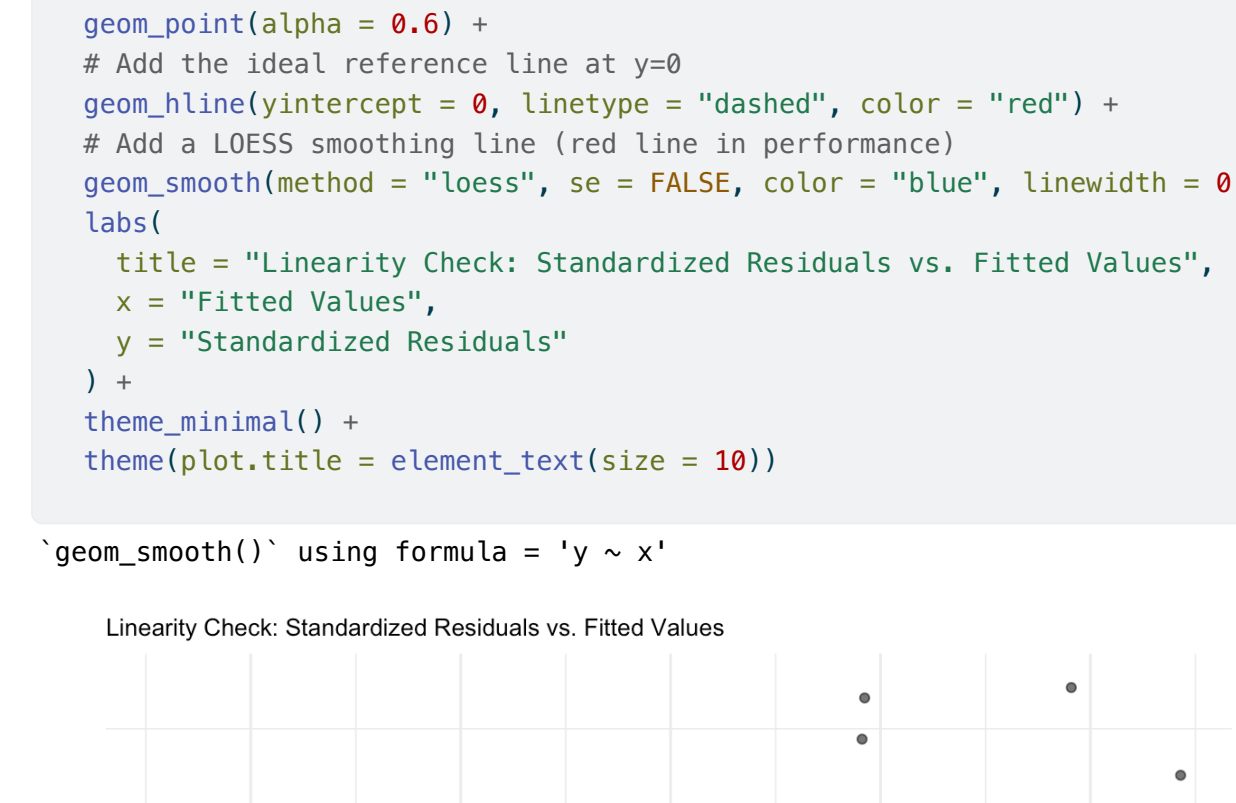
```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: log_Glucose
                          Chisq Df Pr(>Chisq)
(Intercept)          19124.1994  1  < 2.2e-16 ***
SexF                     0.8144  1   0.366829
TimepointF               7.1918  1   0.007324 **
TreatmentF              30.1063  1  4.090e-08 ***
SexF:TimepointF          1.0454  1   0.306574
TimepointF:TreatmentF   16.1239  1  5.933e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Assumptions of the model


# 1. Check the VIF of this model
vif(results_final)
```

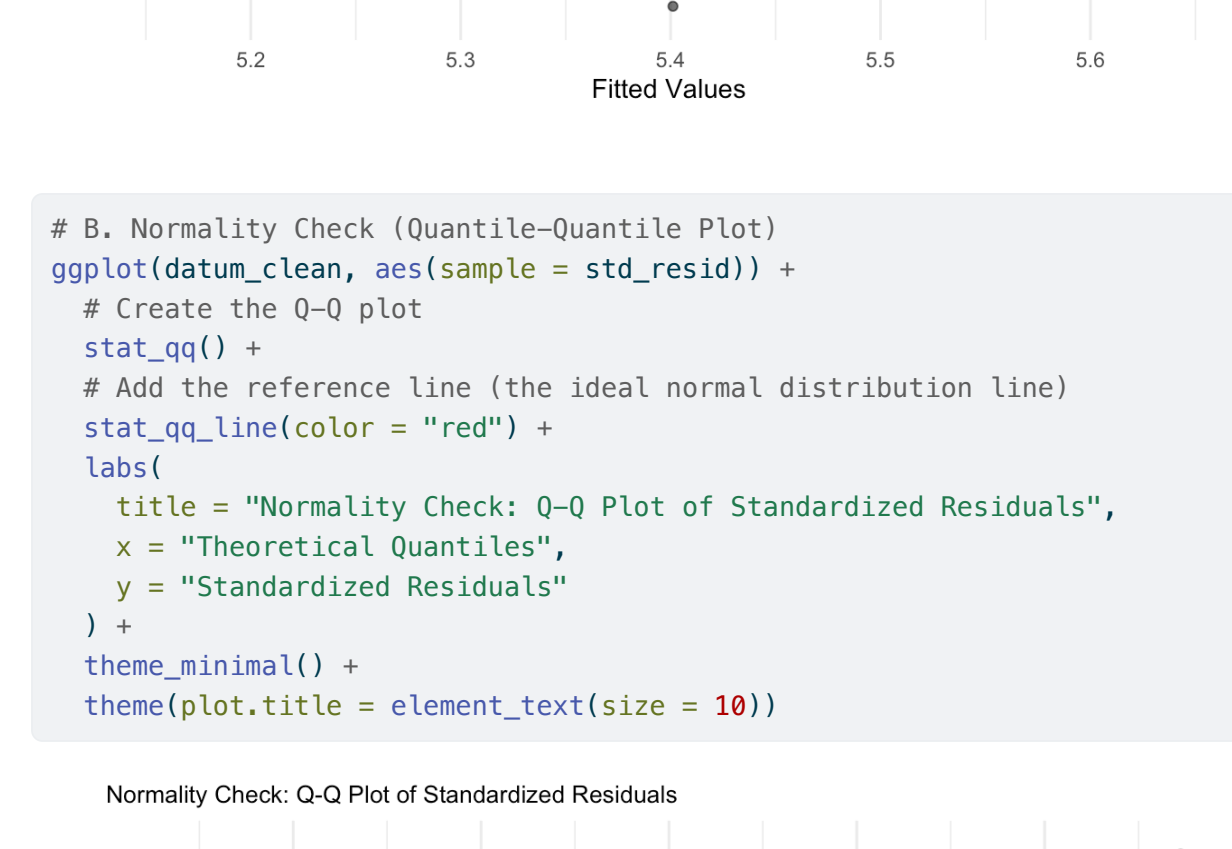
```
                 SexF            TimepointF            TreatmentF
             1.786573              3.029281              1.784715
      SexF:TimepointF TimepointF:TreatmentF
             2.720067              2.881490
```

```
  #VIF = 1: No multicollinearity
  #VIF between 1 and 5: Generally considered acceptable or moderate multicollinearity
  #VIF > 5 (or > 10): Indicates potentially high multicollinearity

# 2. Check other assumptions of a linear model

# --- Extract Residuals and Fitted Values ---
#1. Extract the data frame used by the model
datum_clean <- model.frame(results_final)

# 2. Calculate and assign the fitted and residual values to this clean data frame
datum_clean$fitted <- fitted(results_final)
datum_clean$raw_resid <- residuals(results_final)
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)

# 3. Use the studentized residuals method from the 'lme4' utility functions
# Although 'rstandard' doesn't work, 'residuals' is robust.
# If you really need a standardized residual, use this function:
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)


# --- 2. Create the Plots ---

# A. Linearity and Homoscedasticity Plot (Residuals vs. Fitted)
ggplot(datum_clean, aes(x = fitted, y = std_resid)) +
  geom_point(alpha = 0.6) +
  # Add the ideal reference line at y=0
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  # Add a LOESS smoothing line (red line in performance)
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Linearity Check: Standardized Residuals vs. Fitted Values",
    x = "Fitted Values",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

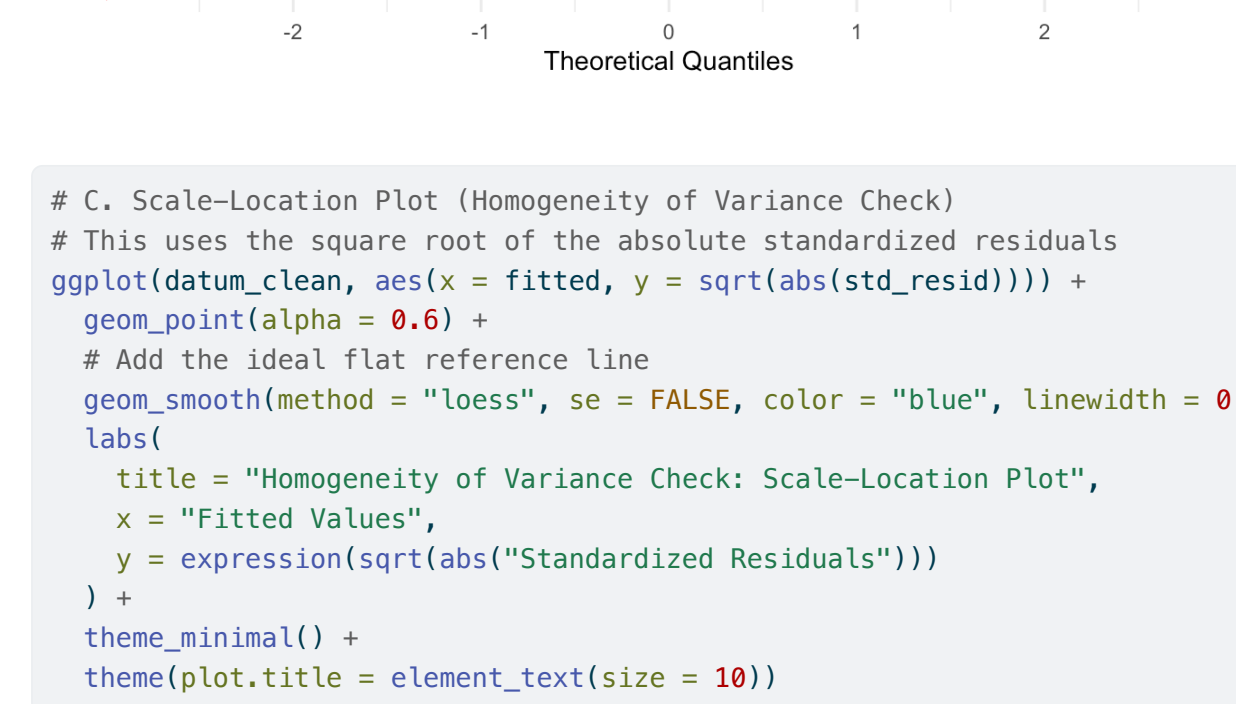


```
# B. Normality Check (Quantile-Quantile Plot)
ggplot(datum_clean, aes(sample = std_resid)) +
  # Create the Q-Q plot
  stat_qq() +
  # Add the reference line (the ideal normal distribution line)
  stat_qq_line(color = "red") +
  labs(
    title = "Normality Check: Q-Q Plot of Standardized Residuals",
    x = "Theoretical Quantiles",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

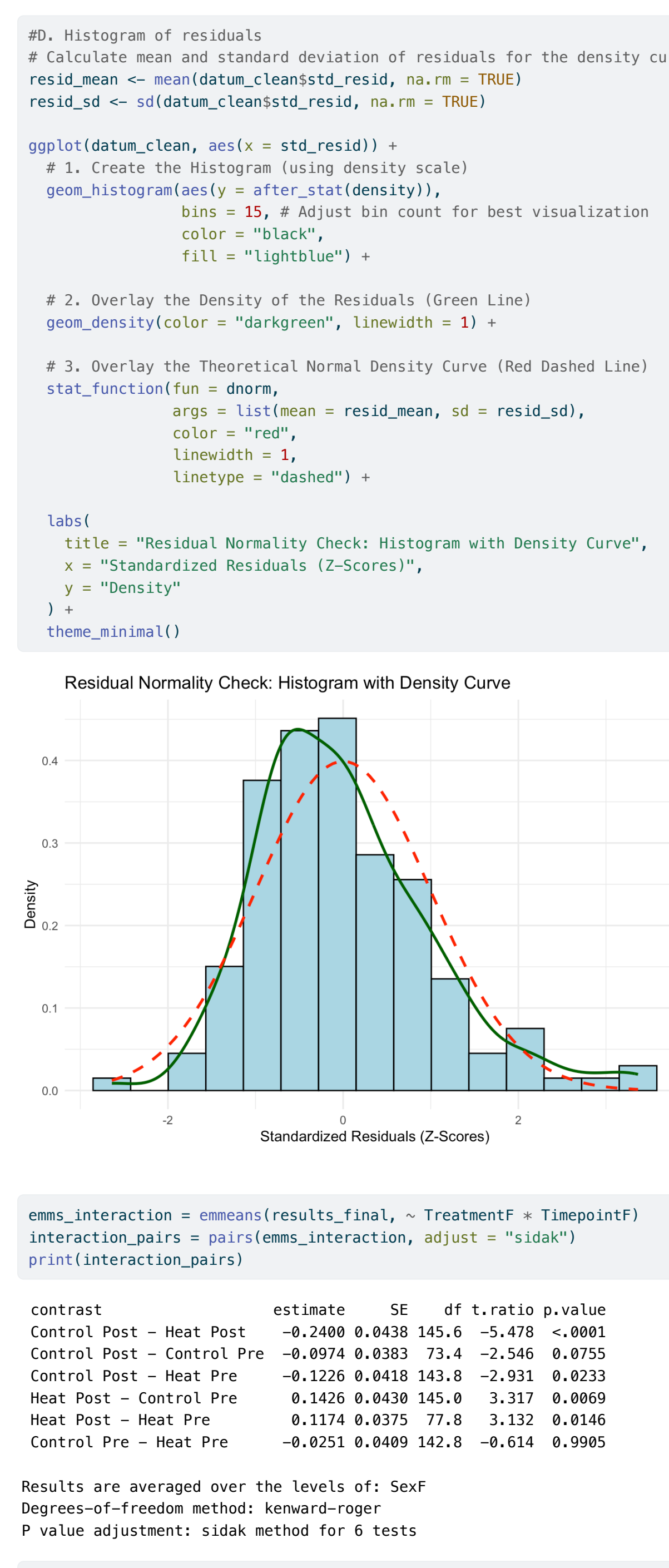


```
# C. Scale-Location Plot (Homogeneity of Variance Check)
# This uses the square root of the absolute standardized residuals
ggplot(datum_clean, aes(x = fitted, y = sqrt(abs(std_resid)))) +
  geom_point(alpha = 0.6) +
  # Add the ideal flat reference line
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Homogeneity of Variance Check: Scale-Location Plot",
    x = "Fitted Values",
    y = expression(sqrt(abs("Standardized Residuals")))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#D. Histogram of residuals
# Calculate mean and standard deviation of residuals for the density curve
resid_mean <- mean(datum_clean$std_resid, na.rm = TRUE)
resid_sd <- sd(datum_clean$std_resid, na.rm = TRUE)

ggplot(datum_clean, aes(x = std_resid)) +
  # 1. Create the Histogram (using density scale)
  geom_histogram(aes(y = after_stat(density)),
                 bins = 15, # Adjust bin count for best visualization
                 color = "black",
                 fill = "lightblue") +

  # 2. Overlay the Density of the Residuals (Green Line)
  geom_density(color = "darkgreen", linewidth = 1) +

  # 3. Overlay the Theoretical Normal Density Curve (Red Dashed Line)
  stat_function(fun = dnorm,
                args = list(mean = resid_mean, sd = resid_sd),
                color = "red",
                linewidth = 1,
                linetype = "dashed") +


  labs(
    title = "Residual Normality Check: Histogram with Density Curve",
    x = "Standardized Residuals (Z-Scores)",
    y = "Density"
  ) +
  theme_minimal()
```



```
emms_interaction = emmeans(results_final, ~ TreatmentF * TimepointF)
interaction_pairs = pairs(emms_interaction, adjust = "sidak")
print(interaction_pairs)
```

```
 contrast                    estimate     SE    df t.ratio p.value
 Control Post - Heat Post     -0.2400 0.0438 145.6  -5.478  <.0001
 Control Post - Control Pre   -0.0974 0.0383  73.4  -2.546  0.0755
 Control Post - Heat Pre      -0.1226 0.0418 143.8  -2.931  0.0233
 Heat Post - Control Pre       0.1426 0.0430 145.0   3.317  0.0069
 Heat Post - Heat Pre          0.1174 0.0375  77.8   3.132  0.0146
 Control Pre - Heat Pre       -0.0251 0.0409 142.8  -0.614  0.9905

Results are averaged over the levels of: SexF
Degrees-of-freedom method: kenward-roger
P value adjustment: sidak method for 6 tests
```

```
cld_output = cld(emms_interaction,
                 adjust = "sidak",
                 Letters = letters)

print(cld_output)
```

```
  TreatmentF TimepointF   emmean         SE       df lower.CL upper.CL .group
1    Control       Post 5.260104 0.03098792 144.5722 5.181942 5.338265    a
3    Control        Pre 5.357517 0.02980037 142.8254 5.282339 5.432695    ab
4       Heat        Pre 5.382659 0.02806716 142.8254 5.311854 5.453464     b
2       Heat       Post 5.500109 0.03097808 146.5505 5.421986 5.578232      c
```

Conclusions: Based on the residuals from the model, this fit is better than the raw data, I still don't love it, but it's what I have used in the past

Next steps: No further steps

Analysis - Body Temperature and Glucose Correlation

```
# Run Pearson's Correlation Test
pearson_result = cor.test(
  x = datum2$PostGlucose,    # Your first continuous variable
  y = datum2$AvgMaxTemp,    # Your second continuous variable
  method = "pearson"   # Specify the Pearson method
)

# View the results
print(pearson_result)
```

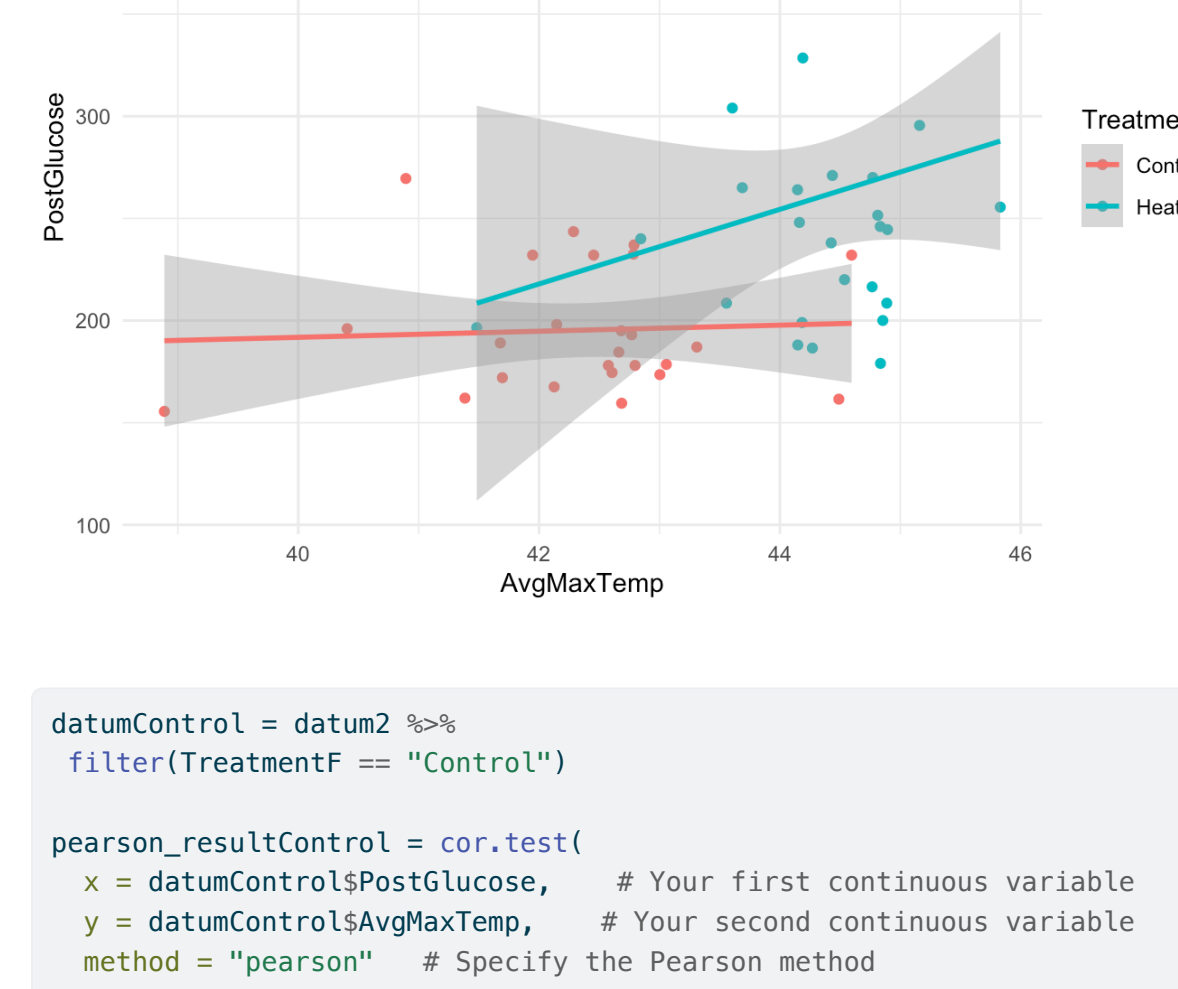
```
Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.4617924

Data:                            datum2$PostGlucose and datum2$AvgMaxTemp

Test Statistic:                  t = 3.71804

Test Statistic Parameter:        df = 51

P-value:                         0.0005001007

95% Confidence Interval:         LCL = 0.2188104
                                 UCL = 0.6508476
```

```
datumHeated = datum2 %>%
 filter(TreatmentF == "Heat")

pearson_resultHeat = cor.test(
  x = datumHeated$PostGlucose,    # Your first continuous variable
  y = datumHeated$AvgMaxTemp,    # Your second continuous variable
  method = "pearson"   # Specify the Pearson method
)
print(pearson_resultHeat)
```

```
Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.2238586

Data:                            datumHeated$PostGlucose and datumHeated$AvgMaxTemp

Test Statistic:                  t = 1.171182

Test Statistic Parameter:        df = 26

P-value:                         0.2521475

95% Confidence Interval:         LCL = -0.1628161
                                 UCL =  0.5509243
```

```
# Visualize the relationship with a scatter plot
ggplot(data = datum2, aes(x = AvgMaxTemp, y = PostGlucose, color = TreatmentF)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) + # 'lm' adds a linear trend line
  labs(title = "Scatter Plot with Linear Regression Line") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
datumControl = datum2 %>%
 filter(TreatmentF == "Control")

pearson_resultControl = cor.test(
  x = datumControl$PostGlucose,    # Your first continuous variable
  y = datumControl$AvgMaxTemp,    # Your second continuous variable
  method = "pearson"   # Specify the Pearson method
)

print(pearson_resultControl)
```

```
Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.05561423

Data:                            datumControl$PostGlucose and datumControl$AvgMaxTemp

Test Statistic:                  t = 0.2671299

Test Statistic Parameter:        df = 23

P-value:                         0.7917498

95% Confidence Interval:         LCL = -0.3471451
                                 UCL =  0.4410530
```

Manuscript Graph

```
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Glucose_Group = mean(Glucose, na.rm = TRUE),
    SD_Glucose_Group = sd(Glucose, na.rm = TRUE),
    SE_Glucose_Group = SD_Glucose_Group / sqrt(N_group),
    CI_Margin_Glucose = 1.96 * SE_Glucose_Group,
    CI_Upper_Group = Mean_Glucose_Group + CI_Margin_Glucose,
    CI_Lower_Group = Mean_Glucose_Group - CI_Margin_Glucose
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 36
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl>   <dbl>
1  4818 M     Control   Poseiden  Right Pre       No    3/3/25   18.0    237
2  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6    198.
3  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1    192.
4  4822 M     Heat      Poseiden  Right Pre       No    3/6/25   16.4    204
5  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4    239
6  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5    226.
# ℹ 26 more variables: Ketones <chr>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, log_Glucose <dbl>, N_group <int>,
#   Mean_Glucose_Group <dbl>, SD_Glucose_Group <dbl>, SE_Glucose_Group <dbl>,
#   CI_Margin_Glucose <dbl>, CI_Upper_Group <dbl>, CI_Lower_Group <dbl>
```

```
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Glucose, color = TreatmentF)) +
  geom_jitter(
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.2, dodge.width = DODGE_WIDTH)
  ) +

  geom_point(
    aes(y = Mean_Glucose_Group, group = TreatmentF),
    size = 7.5,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Glucose_Group, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +


  labs(
    x = "Time Point",
    y = "Corticosterone (ng/mL)",
    color = "Treatment Group"
  ) +
  guides(shape = "none") +

  theme_classic() +

  theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )


# 4. Adding signifigance and sample size

plot1 =  plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(379, 467, 370, 514),
    label = c("a", "a", "a", "b"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
    y = c(379 - 35, 467 - 35, 370 - 35, 514 - 35),
    # Add your sample sizes here
    label = c("(39)", "(36)", "(44)", "(36)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "GlucosePlot.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300 ,
  bg = "transparent"
)
```

Manuscript Graph - With sex

```
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Glucose_Group = mean(Glucose, na.rm = TRUE),
    SD_Glucose_Group = sd(Glucose, na.rm = TRUE),
    SE_Glucose_Group = SD_Glucose_Group / sqrt(N_group),
    CI_Margin_Glucose = 1.96 * SE_Glucose_Group,
    CI_Upper_Group = Mean_Glucose_Group + CI_Margin_Glucose,
    CI_Lower_Group = Mean_Glucose_Group - CI_Margin_Glucose
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 36
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl>   <dbl>
1  4818 M     Control   Poseiden  Right Pre       No    3/3/25   18.0    237
2  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6    198.
3  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1    192.
4  4822 M     Heat      Poseiden  Right Pre       No    3/6/25   16.4    204
5  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4    239
6  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5    226.
# ℹ 26 more variables: Ketones <chr>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, log_Glucose <dbl>, N_group <int>,
#   Mean_Glucose_Group <dbl>, SD_Glucose_Group <dbl>, SE_Glucose_Group <dbl>,
#   CI_Margin_Glucose <dbl>, CI_Upper_Group <dbl>, CI_Lower_Group <dbl>
```

```
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Glucose, color = TreatmentF)) +

  geom_jitter(
  aes(shape = SexF, group = TreatmentF),
  size = 3.5,
  alpha = 0.5,
  position = position_jitterdodge(
    jitter.width = 0.2,
    dodge.width = DODGE_WIDTH
  )
) +

  geom_point(
    aes(y = Mean_Glucose_Group, group = TreatmentF),
    size = 7.5,
    shape = 18,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Glucose_Group, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "Blood Glucose (mg/dL)",
    color = "Treatment Group"
  ) +

  guides(shape = "none") +

  theme_classic() +

  theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )


# 4. Adding signifigance and sample size

plot1 =  plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(379, 467, 370, 514),
    label = c("a", "a", "a", "b"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
    y = c(379 - 35, 467 - 35, 370 - 35, 514 - 35),
    # Add your sample sizes here
    label = c("(39)", "(36)", "(44)", "(36)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "GlucosePlotSex.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300 ,
  bg = "transparent"
)
```